BayesSSD – Bayesian Sample Size Determination for Multilevel Models

Workshop in Maastrich University
24 March 2026

Utrecht University

NWO

Workshop structure

- Theoretical background
- Exercise 1: Modelling Attrition
- Cluster Randomized Trials
- Exercise 2: Sample Size Determination

The Problem

Lack of free software for Bayesian sample size determination for multilevel models

High threshold for using Bayesian hypothesis testing

Bayesian vs. frequentist power

- Frequentist power

	H_0 is true	H_0 is false
H_0 not rejected	✓	Type-II error (β)
H_0 rejected	Type-I error	✓

Bayesian vs. frequentist power

- Frequentist power

	H_0 is true	H_0 is false
H_0 not rejected	✓	Type-II error (β)
H_0 rejected	Type-I error	✓

Power = 1 – P(Type-II error) = 1 – β

Bayesian vs. frequentist power

- Bayesian power
  - Probability of finding a convincing Bayes Factor (BF) in favor of the true hypothesis
  - $BF_{1c} = 2$ ⟹ H_1 receives twice as much support as H_c

	H_1 is true	H_1 is false
BF_{1c} > threshold	✓	Type-I error
BF_{1c} < threshold	Type-II error	✓

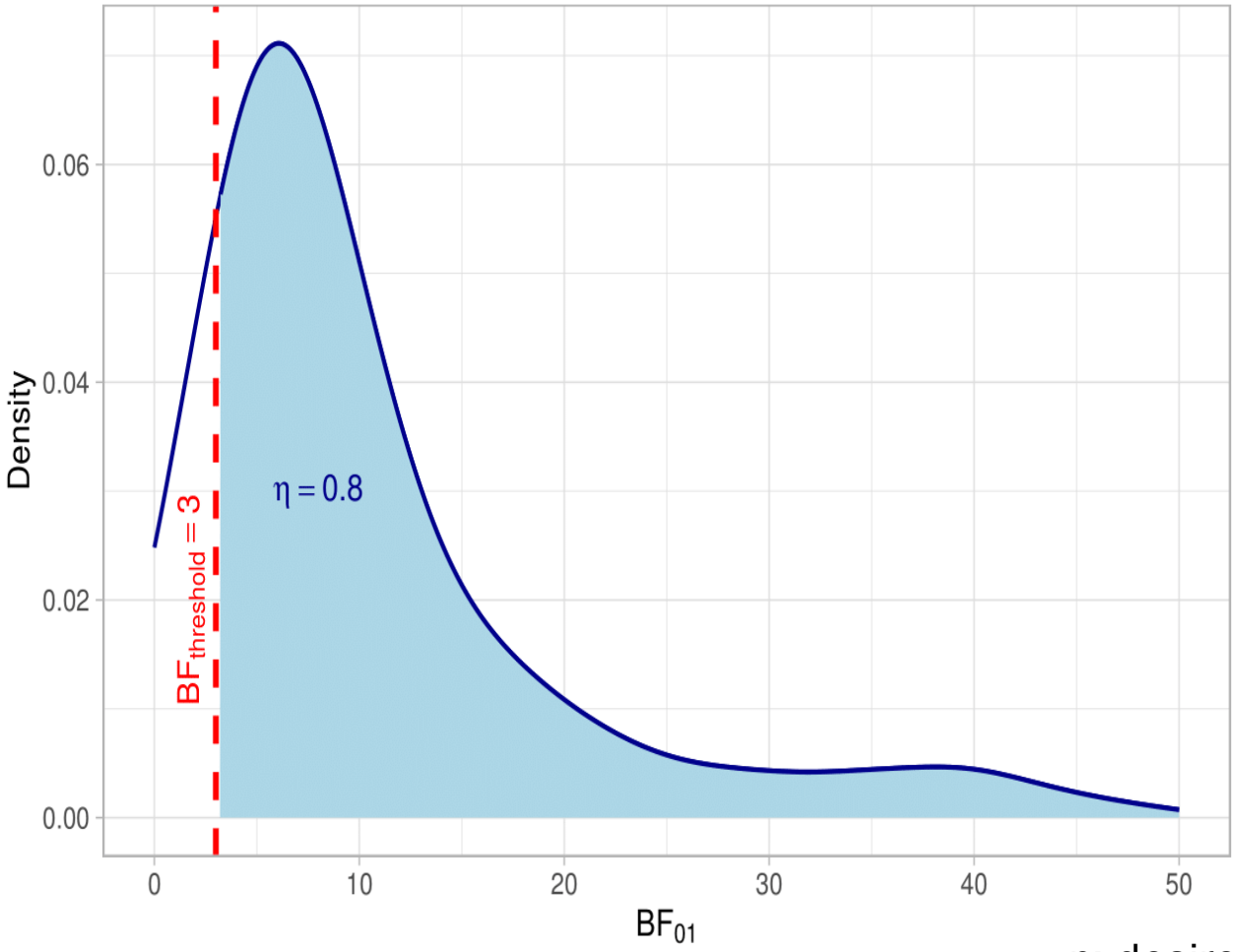
Bayesian vs. frequentist power

- Bayesian power
  - Probability of finding a convincing Bayes Factor (BF) in favor of the true hypothesis
  - $BF_{1c} = 2$ ⟹ H_1 receives twice as much support as H_c
  - Power = $P(BF_{1c} > \text{threshold} \mid H_1)$

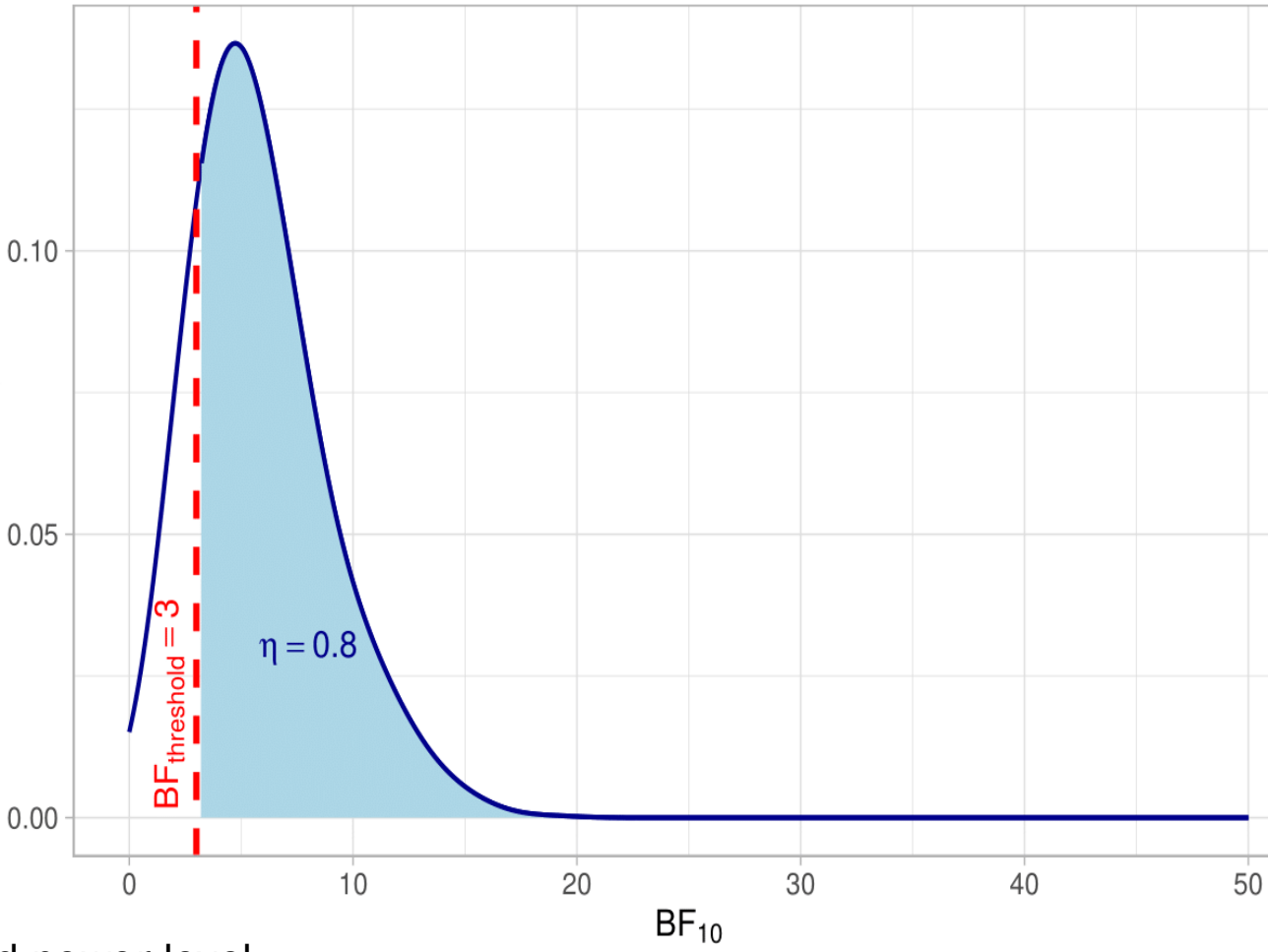
	H_1 is true	H_1 is false
$BF_{1c} > \text{threshold}$	✓	Type-I error
$BF_{1c} < \text{threshold}$	Type-II error	✓

Bayesian vs. frequentist power

H_0=true



H_1=true



η: desired power level

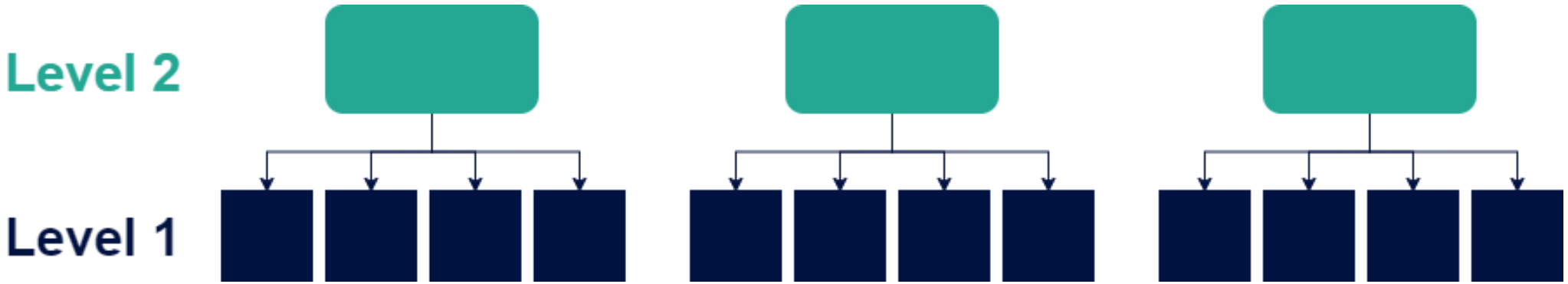
Multilevel Model

- Hierarchical structure
- Correlation of data

$$\beta_{oj} = \gamma_{00} + \gamma_{01} Z_j + u_{oj}$$

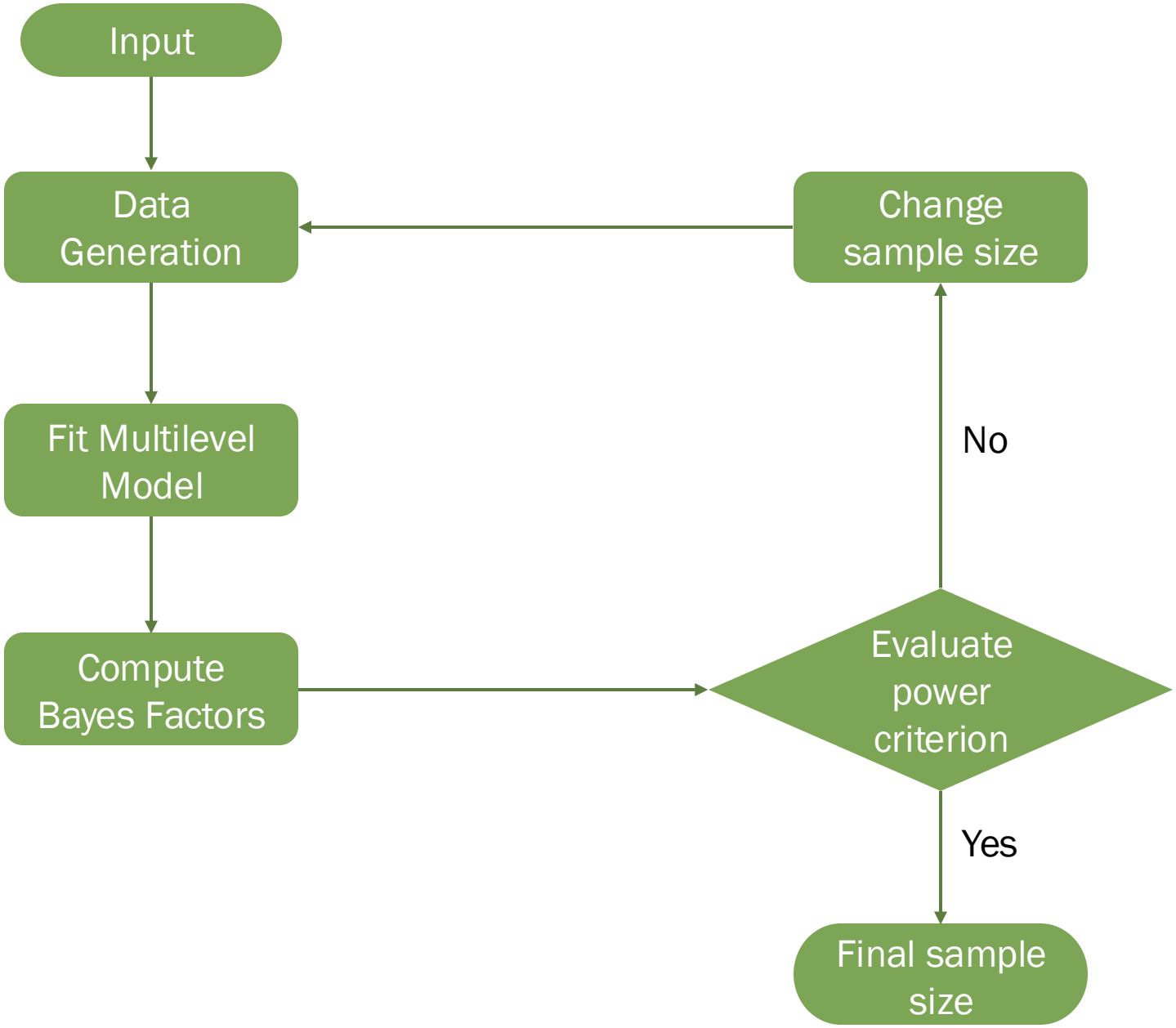
$$\beta_{1j} = \gamma_{10} + \gamma_{11} Z_j + u_{0j}$$

$$y_{ij} = \beta_{oj} + \beta_{1j} X_{ij} + e_{ij}$$



Simulation Algorithm

- Generative model based on input values
- For candidate N, simulate data, fit model, calculate BF, apply decision rule
- Binary search
- Convergence if …



Binary Search

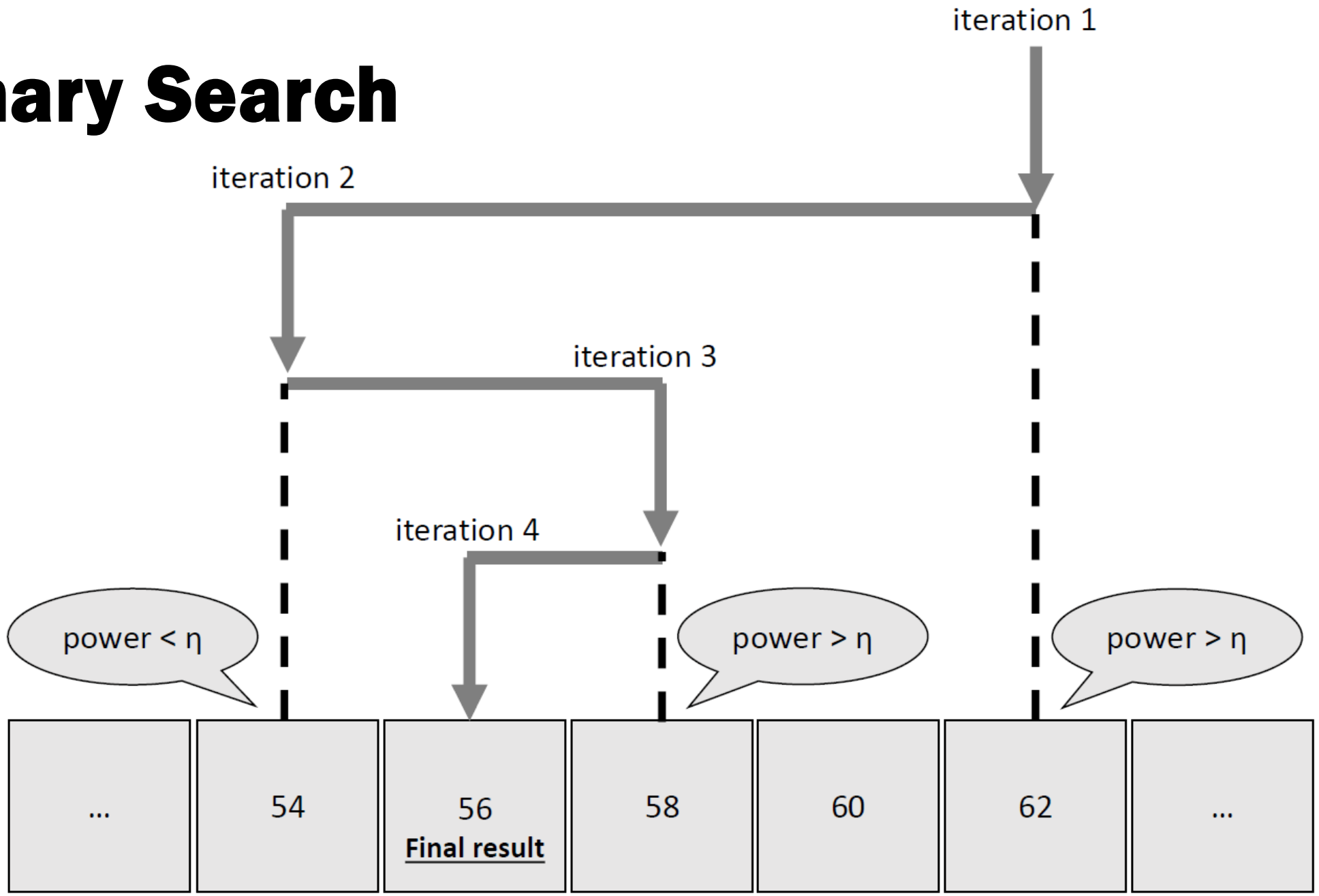


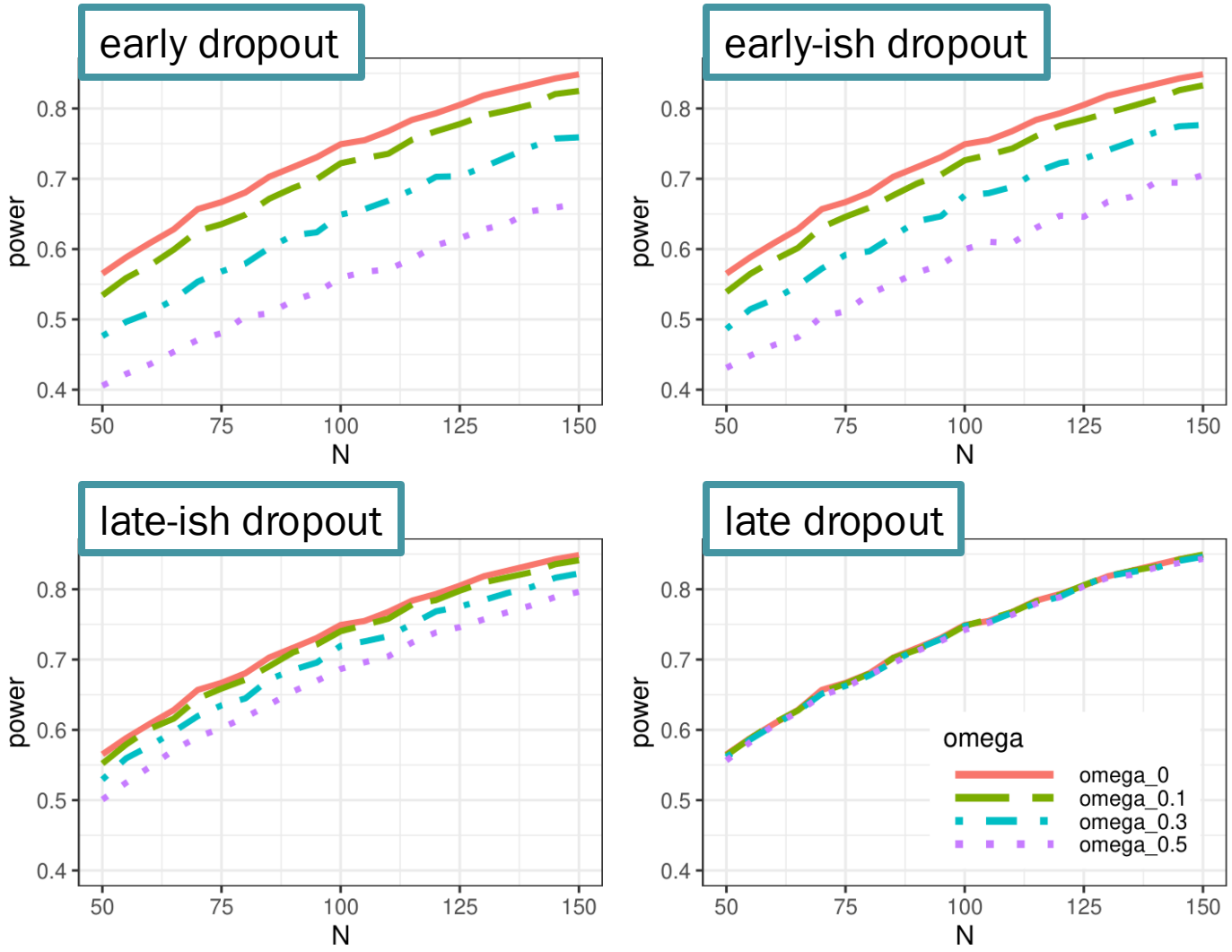
Figure taken from Lösener & Moerbeek, 2025

Attrition and Power

- Attrition negatively impacts power!

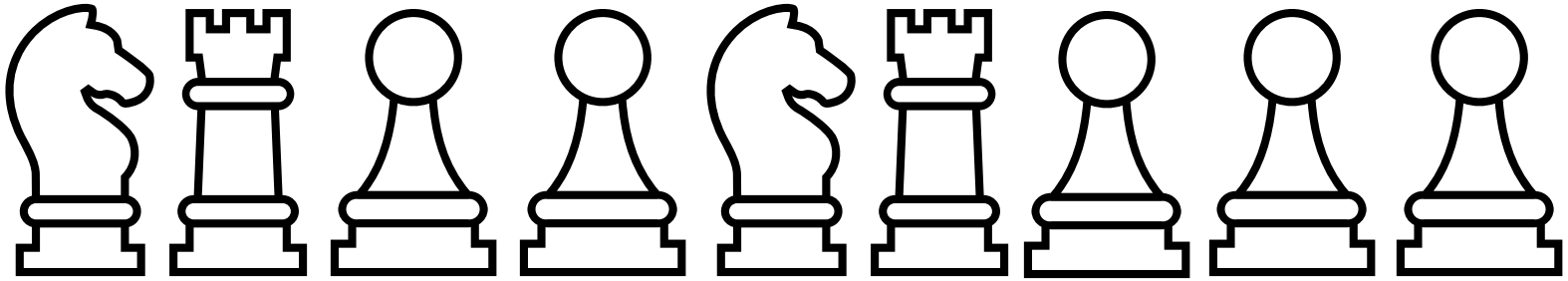
- The *timing* of drop-out matters!

The earlier, the worse

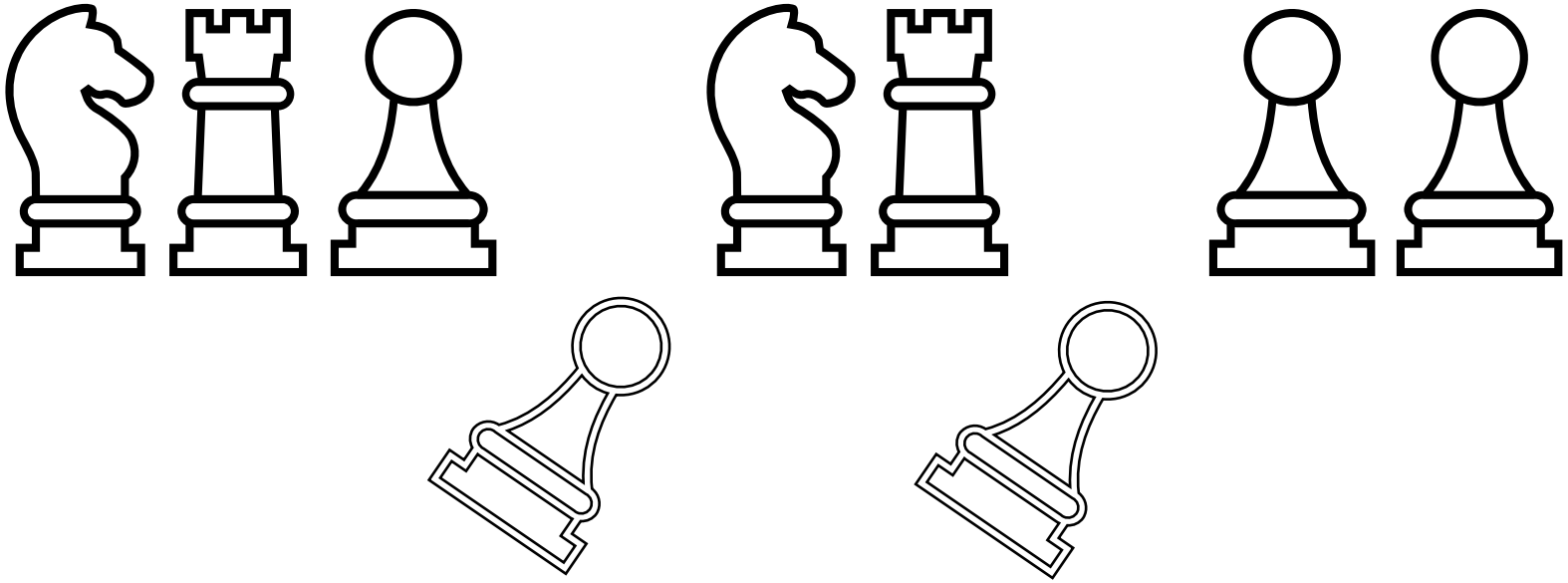


omega: proportion of participants dropping out

Exercise 1 – Model Attrition with Survival Functions



Exercise 1 – Model Attrition with Survival Functions



https://losener.shinyapps.io/attrition-survival/

Attrition | Welcome | Background | Survival by hand | Weibull | Modified-Weibull | Linear-Exponential | Log-Logistic | Gompertz | Non-parametric Survival function | More? | Links

Boring mathematical foundation

Draw your own survival function!

Parametric survival functions

Hazard -> Survival

Survival -> Hazard

This a

What can you do here?

- Explore different parametric and non-parametric survival models
- Adjust parameters and immediately see their effect on the curve
- Draw your own survival curve manually
- See how survival and hazard curves are connected

How to get started

1. Choose a model from the navigation bar
2. Adjust parameters using the input fields
3. Inspect how the survival and hazard functions change
4. Or go to 'Survival by hand' to draw your own curve

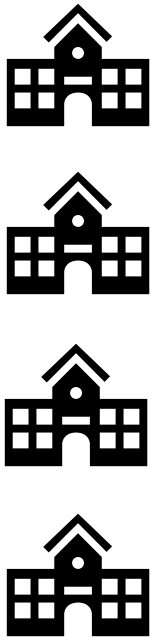
Example: Weibull survival function

- Shows the probability of 'surviving' in the sample up to a certain point
- 80% of participants drop out at some point (omega = .8)

Cluster Randomized Trials

What is it?

Treatment



Control

Sample Size Determination

- Effect size
- Intraclass correlation coefficient (ICC)
- Fixed cluster size or number of cluster
- Bayes factor threshold
- Bayesian power (η)
- Pair of hypotheses

Sample Size Determination

Bayes factor threshold

- Amount of relative support

Bayesian power (η)

- Probability of reaching the threshold

Pair of hypotheses

- What the researcher is testing (null hypothesis, informative hypotheses)

Simulation Results

- Sample size <u>increases</u> with
  - High ICC ⬆
  - Small effect size ⬇
  - High Bayes factor threshold ⬆
  - Including null hypothesis
- It is easier to reach the power criterion when increasing the number of clusters.
- In some scenarios the power criterion was not met after large cluster size (1000).

Package and app

```r
devtools::install_github("ulrichlosener/BayesSSD")


get_sample_size <-SSD_crt_null(eff_size = 0.2,
                               n1 = 10, ndatasets = 1000,
                               rho = 0.05,
                               BF_thresh = 3,
                               eta = 0.8,
                               fixed = "n1",
                               b_fract = 3)
```

```
Final sample size
=================
Hypotheses:
    H0: Intervention=Control
    H1: Intervention>Control
**********************************************************************
     b         n1        n2        P(BF.01 > 3 | H0) P(BF.10 > 3 | H1)
[1,] "  1.000" " 10.000" "104.000" "  0.947"         "  0.802"
[2,] "  2.000" " 10.000" "100.000" "  0.919"         "  0.806"
[3,] "  3.000" " 10.000" " 96.000" "  0.879"         "  0.806"
**********************************************************************
n1: Cluster sizes
n2: Number of clusters
```

https://cnbi.shinyapps.io/appbayessamplsizedet-crt/

Sample Size Determination for Cluster Randomised Trials with Bayes Factor

Results Information

The sample size to be found given the other one is fixed:

- (●) Number of clusters
- ( ) Cluster sizes

Type of hypotheses to compare:

- (●) Null vs. Informative
- ( ) Informative vs. Informative

Cluster sizes: 5

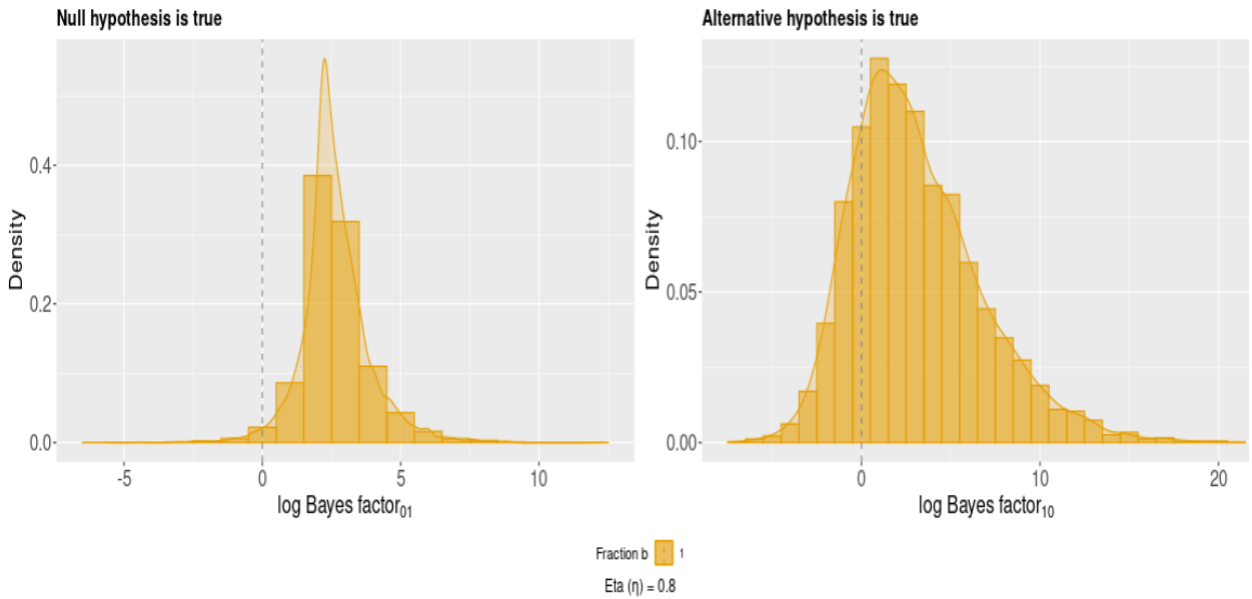
Intraclass correlation: 0.025

Effect size: 0.2

Bayes factor threshold: 1

Fraction b in plots:

- [x] 1
- [ ] 2
- [ ] 3



Final Sample Size

b	Number of clusters	P(BF.01 > thresh)	P(BF.10 > thresh)
1.000	126.000	0.982	0.807
2.000	114.000	0.971	0.801
3.000	108.000	0.963	0.803

Take-home Messages

- There now is an R package for Bayesian SSD in multilevel scenarios!

→ *BayesSSD*

- Future developments: within-interventions, multiple outcomes, contrast testing

- We look forward to your feedback and comments!

References

Gu, X., Mulder, J., & Hoijtink, H. (2018). Approximated adjusted fractional Bayes factors: A general method for testing informative hypotheses. *British Journal of Mathematical and Statistical Psychology, 71*(2), 229-261.

Lösener, U., & Moerbeek, M. (2025). Bayesian sample size determination for longitudinal intervention studies with linear and log-linear growth. *Behavior Research Methods, 57*(9), 239.

Approximate Adjusted Fractional BF

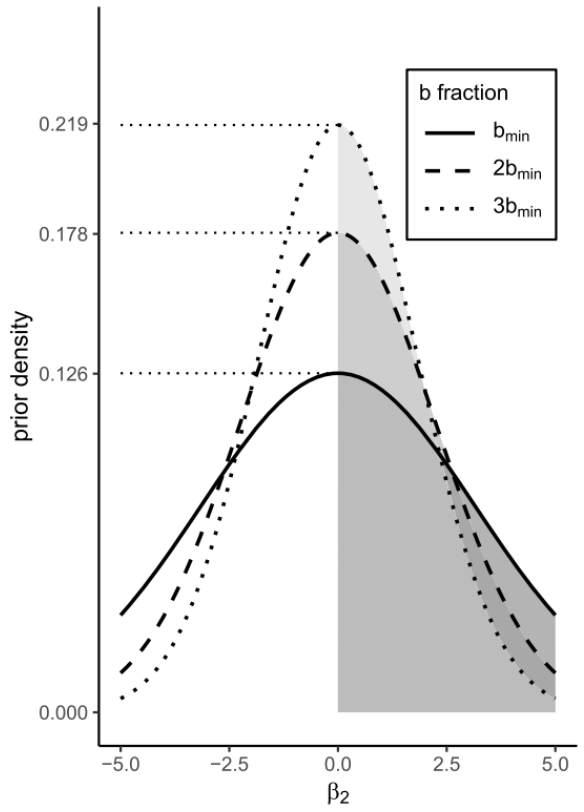
(Gu et al., 2018)

- $BF_{1u} = fit_1 \ / \ complexity_1$
- $BF_{12} = BF_{1u} \ / \ BF_{2u}$

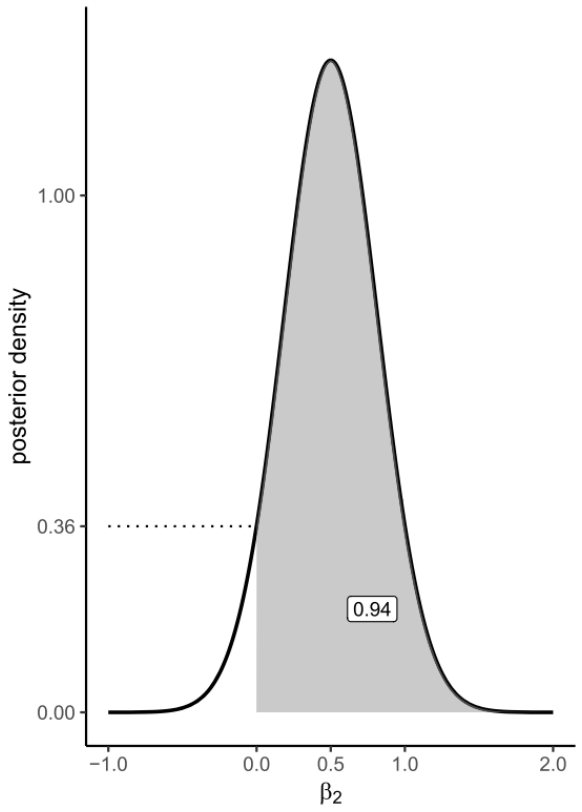
- Default normal prior constructed using fraction of the data (*b*)

b-fraction sensitivity

(see left plot)



(a) *Prior distributions and complexities of $\mathcal{H}_0$ and $\mathcal{H}_1$. The solid, dashed and dotted line corresponds to the prior distribution using b_{min}, $2b_{min}$ and $3b_{min}$, respectively. The horizontal dotted lines indicate the complexity of $\mathcal{H}_0$ under each prior (0.126, 0.178 and 0.219) while the shaded area under the curve indicates the complexity of $\mathcal{H}_1$ under each prior (always 0.5).*



(b) *Posterior distribution for β_2 with $\hat{\beta}_2 = 0.5$ and $\sigma^2_{\hat{\beta}_2} = 0.1$. The fit for $\mathcal{H}_0$ is indicated by the horizontal dotted line and equal to $g_u(\beta_2 = 0 \mid X) = 0.36$. The fit for $\mathcal{H}_1$ is indicated by the shaded area and equal to $g_u(\beta_2 > 0 \mid X) = 0.94$.*

Figure taken from Lösener & Moerbeek, 2025